

Randomized and low-rank approximation

Absolute-error threshold for extremal Gaussian trace bounds

Difficulty: challenging **Importance:** interesting to the community

Status: Solved

Rating rationale: Challenging because signed eigenvalue contributions require uniform extremal tail comparisons; community impact is rigorous trace estimates for indefinite matrices.

Resolution - 2026-09-11

Author: George Stepaniants, Department of Computing and Mathematical Sciences, California Institute of Technology, Pasadena, California, USA.

Solved affirmatively. The [complete proof](#), [Sections 1-8](#) establishes both probability comparisons for every nonzero real symmetric matrix, including indefinite and zero-trace matrices, every positive sample count, and the stated non-strict threshold. Equations (1)-(2) give a stronger normalized one-sided comparison; Section 8 converts it to the exact retained target. [Proof PDF](#) · [Standalone XeLaTeX source](#).

A separate [independent Codex-agent review](#) passed the complete proof and checked Kwaśnicki's published bell-shape theorem and Hallman's coefficient derivative directly. The new inflection identity and grouped transfers are supplied in the manuscript. Substantial ChatGPT/Codex assistance is disclosed; this is independent agent verification, not external human peer review or formal certification. [Submission record and public-status audit](#).

The original target and ID are retained below. The difficulty and importance labels record the historical open question. The auxiliary counterexamples below retain their original author and limited scope.

Related auxiliary counterexamples — 2026-09-11

Author: Matthew J. Colbrook, Department of Applied Mathematics and Theoretical Physics, University of Cambridge. **Independent Codex-agent review:** PASS for the limited result below.

The centered augmented density has a genuine inflection point beyond the proposed auxiliary upper bound. Together with the mode example, this refutes the upper assertions of Hallman Conjectures 1 and 2.

Historical scope: This auxiliary result neither proved nor refuted the complete absolute Gaussian trace-tail probability chain. Failure of an auxiliary sufficient condition does not refute the final tail comparisons. The full target is now resolved by the separate proof above; the auxiliary counterexamples and their attribution remain unchanged.

Primary reference: [complete authored PDF](#), [standalone TeX](#), [Proposition 3.1 \(with Proposition 2.1 for related evidence\)](#). [Independent proof review](#) · [Authorship and submission record](#). Verification is independent agent review, not external human peer review or formal certification.

Problem statement

For a real symmetric $d \times d$ matrix D , define

$$T_m(D) = \frac{1}{m} \sum_{j=1}^m z_j^T D z_j, \quad z_1, \dots, z_m \stackrel{\text{iid}}{\sim} N(0, I_d).$$

Let $A \neq 0$ be an arbitrary real symmetric $n \times n$ matrix, possibly indefinite. Put

$$\lambda = \|A\|_2, \quad \phi = \|A\|_F, \quad \rho = \frac{\phi^2}{\lambda^2}, \quad B_{\lambda, \phi} = \lambda \operatorname{diag}(I_{\lfloor \rho \rfloor}, \sqrt{\rho - \lfloor \rho \rfloor}).$$

The norms are the spectral and Frobenius norms. Let X have Gamma shape $m\rho/2$ and rate $m/(2\lambda)$, so that $\mathbb{E}X = \phi^2/\lambda$. Here a Gamma variable with shape α and rate β has density $\beta^\alpha x^{\alpha-1} e^{-\beta x} / \Gamma(\alpha)$ for $x > 0$.

Conjecture. For every integer $n \geq 1$, every such A , every integer $m \geq 1$, and every

$$\varepsilon \geq \frac{2\lambda}{m} + \sqrt{\frac{2\phi^2}{m} + \left(\frac{2\lambda}{m}\right)^2},$$

the following comparisons hold:

$$\begin{aligned} \Pr(|T_m(A) - \operatorname{tr}(A)| \geq \varepsilon) &\leq 2 \Pr(T_m(B_{\lambda, \phi}) - \operatorname{tr}(B_{\lambda, \phi}) \geq \varepsilon) \\ &\leq 2 \Pr\left(X - \frac{\phi^2}{\lambda} \geq \varepsilon\right). \end{aligned}$$

Zero trailing diagonal entries in $B_{\lambda, \phi}$ are harmless. Each probability uses the appropriate estimator dimension. This is an absolute-error question, even when $\operatorname{tr}(A) = 0$.

Why it matters in numerical linear algebra

The bound would quantify trace-estimation error even when positive and negative eigenvalues cancel.

References

1. Eric Hallman, *Extremal bounds for Gaussian trace estimation*, arXiv:2411.15454v1 (2024), §5, Theorem 7 and Conjecture 4.

Status check

Theorem 7 leaves its threshold unspecified. Checked 2026-09-08: the source still lists only v1. Author, title and 2025/2026 searches, including the [later XTrace paper](#), found no resolution. This is a bounded check.

The displayed conjecture is retained despite a source prose inconsistency: its threshold tends to zero as $m \rightarrow \infty$, whereas the following sentence says it tends to ϕ .

Audit — 2026-09-10

Rechecked [Hallman, Theorem 7 and Conjecture 4](#); the record remains v1. Author and extremal-Gaussian-trace searches found no resolution. The displayed formula matches the numbered conjecture; its inconsistent following prose remains disclosed, and no sharper threshold is claimed proved.